Probabilistic dengue and chikungunya forecasting in Brazil

Team Neural Earth — Methodology, IMDC 2026

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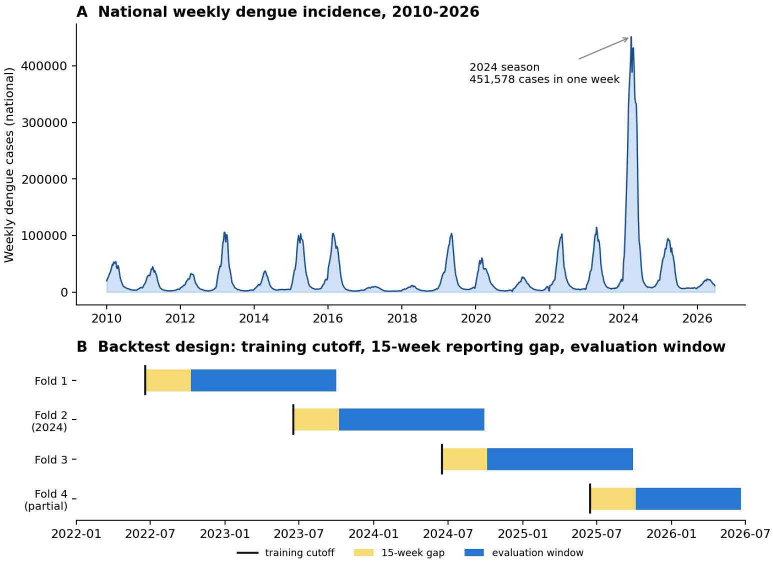
The problem

- Brazil's 2024 dengue season broke the national record by roughly 4$\times$: 451,578 cases in a single epidemiological week, $\sim$6.67M for the year.
- Preparedness needs **calibrated probabilistic forecasts**, not point estimates: the full range of plausible outcomes, not just an expected value.
- Open question this challenge exposes: which modeling paradigm (statistical, ML, DL, mechanistic) forecasts best, and how robustly, when a season can depart by an order of magnitude from the training record?

Our approach: one leakage-safe backtesting harness, five model families, evaluated identically across four official seasons and 26 mandatory states, combined into a conformally-recalibrated ensemble.

Study design



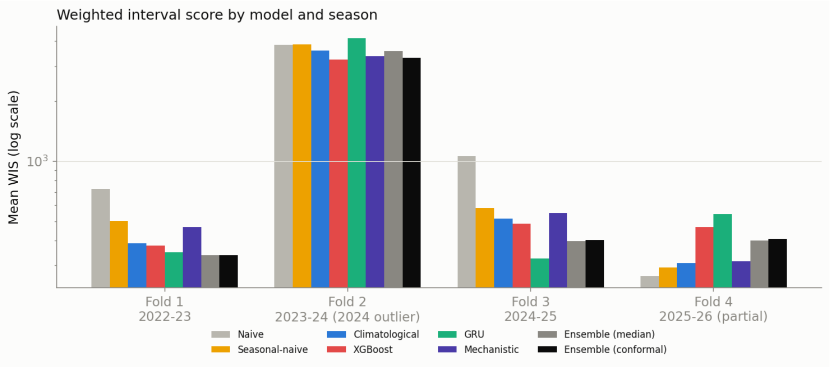
Leakage-safe backtest discipline

- Fold boundaries derived at runtime from the challenge's own `train_N`/`target_N` flags — never hardcoded, so the pipeline stays correct if the data is refreshed.
- A confirmed **15-week reporting gap** sits between each fold's training cutoff and its target window in every official fold. Every joined covariate table (climate, ocean indices, search-access) is explicitly date-filtered to the cutoff, not inferred from the target flag, so this gap cannot leak into training.
- The ECMWF seasonal-forecast product is filtered by *issue date*, not target date: a legitimate future covariate only if it was issued on or before the cutoff.
- A hard leakage assertion guards the whole pipeline, backed by dedicated regression tests.
- Same machinery, unchanged, produces the real forecast-phase submission (train cutoff = EW25 2026, target EW41 2026 $\rightarrow$ EW40 2027).

Five model families, one harness

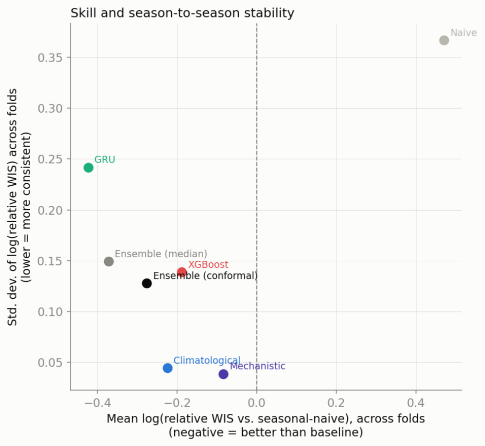
- **Baselines** — naive, seasonal-naive, climatological quantile (empirical same-epiweek quantiles).
- **Gradient boosting** — XGBoost, a single booster with a native multi-quantile objective over all nine levels, on a synthetic-origin panel (weekly origins $\times$ horizons 1–67), with conformalized quantile regression (CQR) calibration. We also tested LightGBM; XGBoost scored significantly better overall and was adopted for dengue. Chikungunya keeps LightGBM, which is the single best model on that target.
- **GRU** — global recurrent network, per-state embeddings, negative-binomial head, 5-member deep ensemble.
- **Mechanistic** — Richards-curve bootstrap of historical trajectories with a negative-binomial observation model; intervals widen with historical variability by construction.
- **Ensemble** — per-quantile median (Vincentization) of climatological + XGBoost + GRU, then conformal recalibration of the tails.

Finding 1: no single model dominates



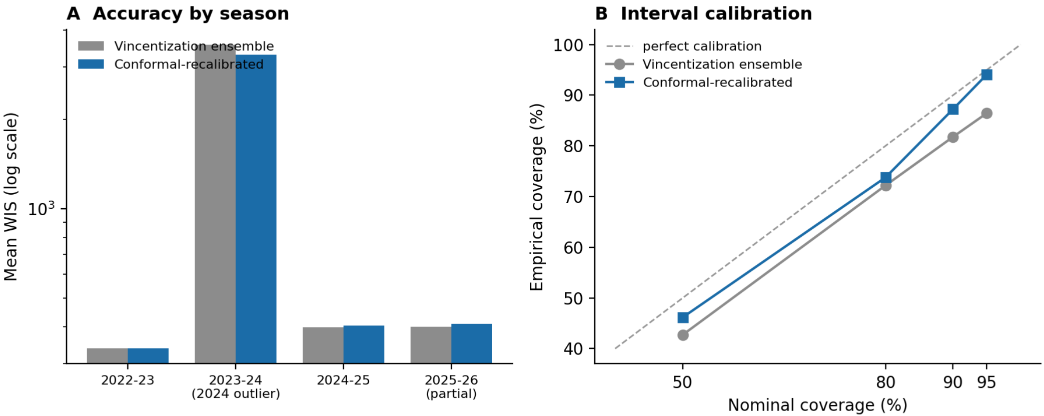
The GRU is best in ordinary seasons but worst on the 2024 outlier. XGBoost is strongest on the outlier and, unlike the LightGBM alternative we also tested (which collapsed on the ordinary seasons), stays reasonably competitive there too. The ensemble never suffers a member's worst-season failure.

Skill and stability are different axes



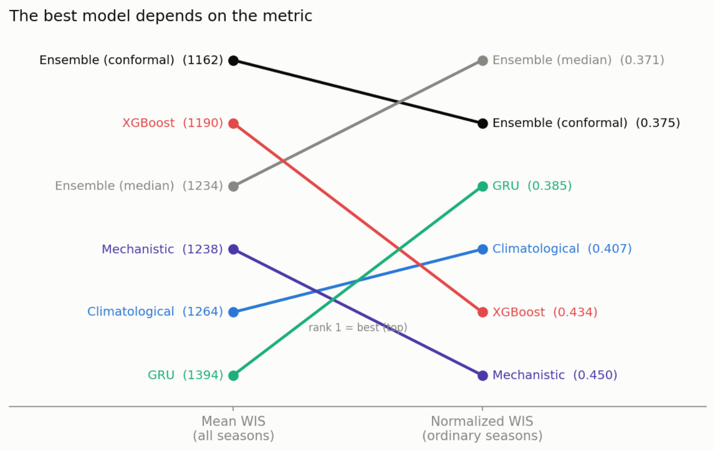
Mean vs. std-dev of log relative WIS across folds. The GRU has the best mean skill but is among the least stable; the mechanistic model is the most stable but modest; the ensembles occupy a position no single member reaches.

Finding 2: conformal recalibration is cheap, transferable insurance



Factors $\{50\text{: } 1.08, 80\text{: } 1.04, 90\text{: } 1.19, 95\text{: } 1.86\}$ tuned on one season, applied to all others. Mean WIS 1234 $\rightarrow$ 1162 (out-of-sample -6.2%); coverage 43/72/82/86 $\rightarrow$ 46/74/87/94. Gain concentrates in the outbreak season at negligible cost elsewhere.

Finding 3: the winner depends on the metric

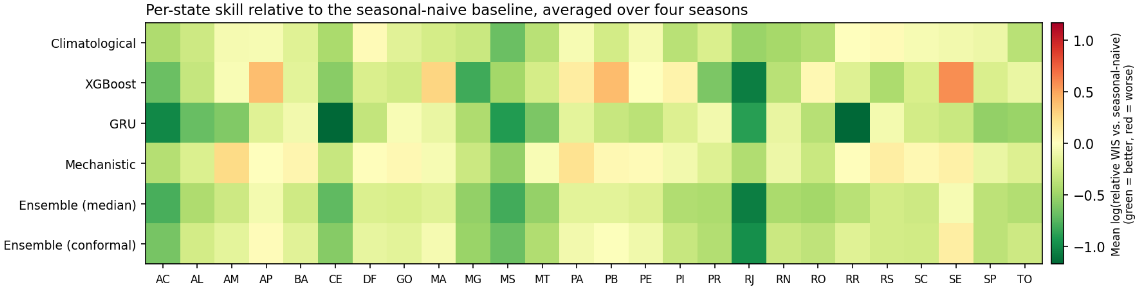


Magnitude-weighted mean WIS and scale-free normalized WIS reward accuracy on different subsets of a highly skewed set of states. We report both; a single aggregate can mislead.

Finding 4: added complexity did not pay

- **Observed climate covariates** added to the gradient-boosted model: WIS **worsened** ($1166 \to 1190$, -2.0%; replicates the original LightGBM-era result of $1299 \to 1342$, -3.3%). The forecast-relevant signal is the ECMWF seasonal forecast and ENSO, not observed weather — observed climate is already implicit in recent case lags.
- **Hierarchical macroregion pooling** of seasonal climatology: WIS worsened ($1162 \to 1185$). States are too heterogeneous within a macroregion for naive pooling to help.
- **Hyperparameter tuning, re-examined:** the original LightGBM-era result was a caution (holdout improved, true backtest worsened). Re-verified on XGBoost, it *reverses* standalone — the holdout's shallower config wins the true backtest too (-3.2%) — but *still fails* the same ensemble-level fold-1 discipline that validated every adopted change. Not adopted either way: a caution that even a validation rule's own conclusions can be implementation-specific.
- Simple methods (climatological quantile, a historical-trajectory bootstrap) are hard to beat, consistent with small-sample, few-series forecasting-competition evidence.

Beyond the aggregate: per-state and disease-specific results



Per-state skill relative to the seasonal-naive baseline. The GRU's advantage concentrates in specific states rather than being uniform. Chikungunya: gradient boosting alone beats the ensemble (WIS 79.1 vs. 89.5), in every season — the best model is disease-specific.

Independent recognition and current status

- At the 31 July 2026 validation-round webinar, our submission was named:
  - one of **6 best-overall-performing models** (of 27) for the mandatory dengue state-level target;
  - one of **3 models with outstanding performance** on the optional city-level target.
- **Forecast phase (2026–27 season) submitted**, ahead of the 10 September deadline: generated 7 September on data refreshed through EW25 2026, then regenerated 9 September to use the now-adopted XGBoost ensemble member and a zero-median edge-case fix (from earlier validation-round organizer feedback) not yet wired into the forecast script.
- Public repository: leakage tests, scoring-correctness tests, determinism guards, provenance manifest, and validated submissions for all four official tracks.

Takeaways

1. Model rankings are season-dependent; a model chosen on average performance can be the least reliable exactly when an outbreak makes accuracy matter most.
2. An unweighted ensemble is never the single worst model in any season, and conformal recalibration turns it into the single best, at negligible cost.
3. Metric choice is a reporting-standards issue, not a modeling one: report both magnitude-weighted and scale-free skill.
4. Complexity did not pay here: the signal that helps is the *right* covariate (seasonal forecast, not observed weather), not more of it.

`github.com/kamrul28890/3rd_imdc_purdue_neuralearth`